

Notes: Phelps (AER, 1972)

The Statistical Theory of Racism and Sexism

Contents

1	One-sentence summary	2
2	Big picture	2
3	Incremental contribution of the paper	2
3.1	A formal model of statistical discrimination	2
3.2	Separating prejudice from inference	3
3.3	A clean bridge from information economics to labor-market inequality	3
3.4	Group means and group variances both matter	3
3.5	Foundation for later work	3
4	Conceptual framework	3
4.1	Information scarcity in the labor market	3
4.2	Employer objective	3
4.3	Group identity as a proxy	3
4.4	Why the model is not a defense of discrimination	4
5	Model setup and derivations	4
5.1	True qualification and noisy test score	4
5.2	Case with no group information	4
5.3	Adding group identity	4
5.4	Prediction using test score and group identity	5
6	Main cases in the paper	5
6.1	Case 1: lower perceived group mean and equal variance	5
6.2	Case 2: group-specific variance in latent promise	5
6.3	Crossing prediction curves	6
6.4	Further case: group differences in test reliability	6
7	Economic intuition	6
7.1	The signal extraction problem	6
7.2	Why group identity affects the prediction	6
7.3	Why this can be self-perpetuating	6
7.4	Why the paper matters for policy	7
8	Identification and interpretation	7
8.1	What the paper identifies theoretically	7
8.2	Observable implications	7
8.3	Important limitations	7
9	Tables and figures	7
9.1	Figure 1: Prediction of Qualification by Race and Test Score	8
10	Short summary	9

11 Conclusion	9
11.1 Core conclusion	9
11.2 Economic intuition	9
11.3 Incremental contribution revisited	9
11.4 Policy implication	10
11.5 Limitations	10
11.6 Takeaway	10

1 One-sentence summary

Phelps develops a compact statistical-discrimination model showing how an employer who has no taste-based prejudice can still rationally use race or sex as a proxy for missing information when applicant quality is imperfectly observed; the paper explains how group-level priors and perceived signal reliability can generate different hiring predictions for applicants with the same test score.

2 Big picture

- **Question.** How can discrimination arise when employers are not motivated by animus, but instead are trying to maximize expected profit under imperfect information?
- **Core idea.** Employers observe a noisy measure of an applicant’s true qualification. If group identity is observed and believed to be statistically informative about either average qualification or signal reliability, the employer combines the test score with the group signal.
- **Difference from taste discrimination.** Becker-style taste discrimination is driven by a dislike of working with or hiring a group. Phelps’ model instead studies “liberal” employers who have no such distaste but use group membership as a statistical proxy.
- **Bayesian logic.** The employer forms a posterior prediction of the worker’s qualification by weighting the noisy individual signal and the group-level prior. More reliable information receives more weight.
- **Discrimination channel.** If an employer believes that one group has a lower expected qualification because of social disadvantage or past experience, then two applicants with the same test score may receive different predicted qualifications.
- **Variance channel.** If the employer believes that the disadvantaged group has a wider distribution of latent promise, the same test score may be interpreted differently. High scores by members of the disadvantaged group may receive more weight because they are seen as more informative.
- **Signal-reliability channel.** If the employer believes that test scores for one group are less reliable, the test score receives less weight for that group. This can reverse or modify the slope patterns in the prediction curves.
- **Economic takeaway.** Statistical discrimination can be individually rational for the employer but socially damaging and potentially self-perpetuating for the targeted group.

3 Incremental contribution of the paper

3.1 A formal model of statistical discrimination

The paper gives one of the earliest exact models of statistical discrimination in labor markets. Instead of treating discrimination as purely taste-based, it formalizes a setting in which the employer has imperfect information and uses observable group identity as a proxy for unobserved worker characteristics.

3.2 Separating prejudice from inference

The model clarifies that discriminatory outcomes do not require employers to dislike minorities or women. The employer can be “liberal” in the sense of having no direct disutility from hiring members of a group, while still making group-contingent predictions because information is incomplete.

3.3 A clean bridge from information economics to labor-market inequality

The paper links labor-market discrimination to the general problem of costly information. The restaurant analogy in the paper emphasizes that using categories as proxies may be privately rational when individual-specific information is expensive, even though the rule can be wrong in many individual cases.

3.4 Group means and group variances both matter

The paper’s contribution is not only that disadvantaged groups may be assigned lower predicted qualifications because of lower perceived means. It also shows that perceived group variance and perceived test-score reliability affect the slope of the prediction rule. Thus discrimination can arise through both levels and information precision.

3.5 Foundation for later work

The model provides a building block for later statistical-discrimination theories, including models where beliefs become self-confirming. In later papers such as Coate and Loury, worker investment decisions respond to employer beliefs, making stereotypes endogenous and persistent.

4 Conceptual framework

4.1 Information scarcity in the labor market

The starting point is that the labor market is not Walrasian in the sense of perfect, costless information. Employers do not directly observe all relevant applicant characteristics. They may observe test scores, resumes, interviews, and group identity, but these signals are imperfect.

4.2 Employer objective

The employer wants to hire or assign workers based on expected productivity. The employer’s decision is therefore based not on true qualification q_i , which is unobserved, but on a prediction of q_i constructed from available information.

4.3 Group identity as a proxy

Skin color or sex can become an informational proxy when the employer believes that group membership is statistically related to qualification or to the reliability of observed signals. The paper emphasizes two possible sources of such beliefs:

- **Statistical experience.** Employers may have observed past differences in average outcomes across groups.

- **Sociological beliefs.** Employers may believe that social disadvantage causes lower average preparation for a targeted group.

This is the core of statistical discrimination: the employer uses a group characteristic to fill in missing information about the individual.

4.4 Why the model is not a defense of discrimination

The paper explicitly states that discrimination is no less damaging if it is statistical. The model is an explanation of one possible mechanism, not a claim that discrimination is harmless or normatively acceptable.

5 Model setup and derivations

5.1 True qualification and noisy test score

Let q_i denote applicant i 's true qualification or promise. The employer observes a test score y_i that equals true qualification plus noise:

$$y_i = q_i + \mu_i, \quad (1)$$

where μ_i is normally distributed with mean zero.

Interpretation. The test score is informative but imperfect. A high score tends to indicate high qualification, but some of the score reflects noise.

5.2 Case with no group information

If the employer observes no group information, the best linear prediction of qualification is based only on the test score. In deviation-from-mean form, Phelps writes:

$$q'_i = a_1 y'_i + u_i, \quad (2)$$

with

$$0 < a_1 = \frac{\text{var}(q')}{\text{var}(q') + \text{var}(\mu_i)} < 1, \quad E[u_i] = 0. \quad (3)$$

Interpretation. The slope a_1 is a reliability weight. If the test score is very noisy, $\text{var}(\mu_i)$ is large and a_1 is small, so the employer shrinks the prediction strongly toward the mean. If the test is precise, a_1 is close to one, so the employer relies heavily on the test score.

5.3 Adding group identity

Now suppose the employer observes whether an applicant belongs to a group, such as whether the applicant is Black or white in the paper's notation. Let $c_i = 1$ if the applicant is Black and $c_i = 0$ otherwise. The employer postulates that true qualification is

$$q_i = \alpha + x_i + \eta_i, \quad (4)$$

where x_i captures group-related social factors and η_i is individual variation. Phelps writes the group-related component as

$$x_i = (-\beta + \varepsilon_i)c_i, \quad \beta > 0. \quad (5)$$

Interpretation. The term $-\beta c_i$ captures a perceived negative group mean effect. The term $\varepsilon_i c_i$ allows for additional group-specific dispersion in social factors.

Define

$$\lambda_i = \eta_i + c_i \varepsilon_i, \quad z_i = -\beta c_i. \quad (6)$$

Then

$$q_i = \alpha + z_i + \lambda_i, \quad (7)$$

while the observed test score is

$$y_i = q_i + \mu_i = \alpha + z_i + \lambda_i + \mu_i. \quad (8)$$

5.4 Prediction using test score and group identity

The employer can now use both the test score and group identity. In deviation form, the prediction of qualification net of the group component is

$$q'_i - z'_i = a_1(y'_i - z'_i) + u_i, \quad (9)$$

where

$$0 < a_1 = \frac{\text{var}(\lambda_i)}{\text{var}(\lambda_i) + \text{var}(\mu_i)} < 1. \quad (10)$$

Equivalently,

$$q'_i = \frac{\text{var}(\lambda_i)}{\text{var}(\lambda_i) + \text{var}(\mu_i)} y'_i + \frac{\text{var}(\mu_i)}{\text{var}(\lambda_i) + \text{var}(\mu_i)} z'_i + u_i. \quad (11)$$

Interpretation. The predicted qualification is a weighted average of the individual test signal and the group-information component. The relative weights depend on variance. If the test is noisy, the employer places more weight on the group prior. If the latent individual component is highly variable, the employer places more weight on the test score.

6 Main cases in the paper

6.1 Case 1: lower perceived group mean and equal variance

If the employer believes that growing up Black is socially disadvantageous, then $z'_i < 0$ for Black applicants. If there is no differential variability in promise across groups, then the prediction curve for Black applicants is parallel to and below the prediction curve for white applicants.

Implication. Two applicants with the same test score can receive different predicted qualifications. The difference comes from the prior mean, not from different test-score slopes.

6.2 Case 2: group-specific variance in latent promise

In the general model, the variance of the latent component λ_i depends on group identity:

$$\text{var}(\lambda_i) = \text{var}(\eta_i) + c_i^2 \text{var}(\varepsilon_i). \quad (12)$$

For Black applicants, $c_i = 1$, so their latent variance is larger if $\text{var}(\varepsilon_i) > 0$. For white applicants, $c_i = 0$, so the additional variance term drops out.

Implication. The coefficient on the test score is larger for Black applicants than for white applicants. The employer sees a high test score from a Black applicant as especially informative because it comes from a group the employer believes has a wider distribution of underlying promise. In the limit, as the group-specific variance becomes very large, the slope for Black applicants approaches one.

6.3 Crossing prediction curves

Because the slope can be steeper for the disadvantaged group, the prediction curves can cross. At low test scores, the lower group mean can dominate, so the Black applicant is predicted to have lower qualification. At sufficiently high test scores, the steeper slope can dominate, so the Black applicant may be predicted to exceed a white applicant with the same or even a lower score.

Economic message. Statistical discrimination does not always imply a uniform penalty at every score. It can imply nonlinear or score-dependent treatment because employers combine group priors and individual evidence.

6.4 Further case: group differences in test reliability

Phelps also considers the possibility that the test-score error term depends on race:

$$\mu_i = \xi_i + c_i \rho_i. \quad (13)$$

If the employer believes that test scores for Black applicants are less reliable than those for white applicants, the employer places less weight on Black applicants' test scores. This can offset or reverse the steeper-slope effect from Case 2.

Implication. Differences in the perceived reliability of signals are central. Even if two applicants have identical observed scores, the score may be treated as more or less informative depending on the employer's beliefs about measurement error.

7 Economic intuition

7.1 The signal extraction problem

The model is a signal-extraction problem. The employer sees a noisy signal and forms a posterior prediction of true productivity. The prediction rule combines:

- the applicant's observed test score;
- the employer's group-level prior;
- the perceived reliability of the test score;
- the perceived variability of true qualification within each group.

7.2 Why group identity affects the prediction

Group identity affects the prediction because the employer believes that it shifts either the expected value or the distribution of qualification. If group identity were completely uninformative, it would not enter the prediction rule. But if the employer believes it is informative, then group-contingent predictions follow mechanically from least-squares/Bayesian updating.

7.3 Why this can be self-perpetuating

The paper notes that group priors may come from sociological beliefs about disadvantage. If discrimination itself creates disadvantage - for example by reducing opportunities, incentives, or experience - then statistical discrimination can become self-perpetuating. This point becomes central in later models of self-fulfilling stereotypes.

7.4 Why the paper matters for policy

The final paragraph emphasizes that statistical discrimination is harmful even if it is not motivated by hatred. A policy response is still important because workers can be harmed by group-based inference even when employers believe they are using information rationally.

8 Identification and interpretation

8.1 What the paper identifies theoretically

The paper does not estimate a model empirically. It provides a theoretical explanation for how discriminatory treatment can arise from incomplete information. The key object is the employer's prediction rule, not an empirical causal estimate.

8.2 Observable implications

The model implies several qualitative patterns:

- workers from a disadvantaged group may receive lower predicted qualifications for the same test score;
- the slope relating test scores to predicted qualifications may differ across groups;
- group differences in test reliability can alter or reverse the slope differences;
- discrimination may persist even without taste-based prejudice.

8.3 Important limitations

The model treats employer beliefs as given rather than fully explaining how they are formed. It also abstracts from endogenous worker investment, institutional constraints, employer learning, strategic responses, and general-equilibrium labor-market effects. These limitations are understandable given the paper's purpose: to introduce the statistical-discrimination mechanism in a simple, exact framework.

9 Tables and figures

The paper contains one figure and no tables. Following the requested format, the figure name appears as a subsection title, and the interpretation is placed immediately after the figure.

9.1 Figure 1: Prediction of Qualification by Race and Test Score

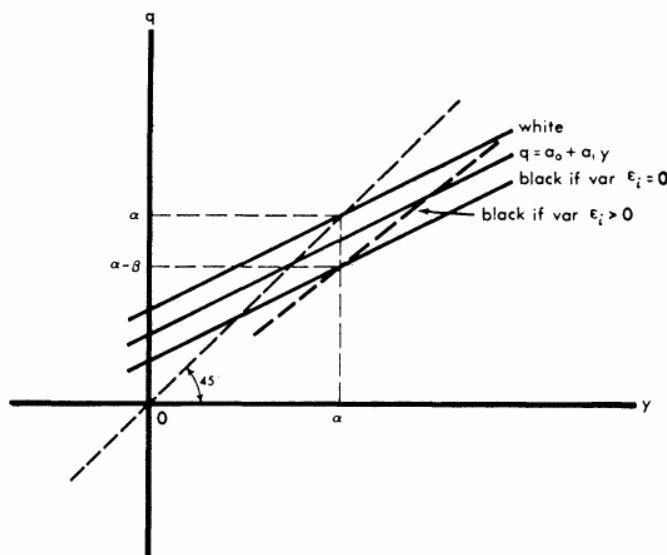


FIGURE 1. PREDICTION OF QUALIFICATION
BY RACE AND TEST SCORE

Figure 1: Prediction of Qualification by Race and Test Score. Prediction curves for qualification under different assumptions about group means, group variance, and test-score reliability.

Read:

- The horizontal axis is the observed test score y and the vertical axis is predicted qualification q .
- The white prediction curve is labeled $q = \alpha_0 + a_1 y$.
- When the employer believes that Black applicants have the same latent variance as white applicants but a lower group mean, the Black prediction curve is parallel to and below the white curve.
- When the employer believes that Black applicants have greater latent variance, the Black prediction curve becomes steeper. At sufficiently high scores, it can cross the white curve.
- The dashed 45° line helps show the benchmark where predicted qualification equals the test score.

Interpretation: The figure condenses the paper's main logic. Group membership affects the intercept when the employer believes the group's mean preparation is lower. Group membership affects the slope when the employer believes the group's latent variance or signal reliability differs. Thus, statistical discrimination can appear as a level penalty, as a different response to scores, or as both.

Economic message: The same test score need not produce the same predicted productivity across groups when employers use group identity as a proxy for missing information. The figure also shows that the effect can vary over the score distribution: the disadvantaged group may be

penalized at low or moderate scores but not necessarily at very high scores if high scores are viewed as especially informative.

10 Short summary

- Phelps provides a compact statistical theory of racial and sexual discrimination.
- The model differs from taste-based discrimination: employers need not dislike minority or female workers.
- Employers observe noisy test scores and may observe group identity.
- If group identity is believed to be informative about average qualification or signal reliability, it enters the employer's prediction of true productivity.
- When the disadvantaged group is believed to have a lower mean qualification, the prediction curve for that group lies below the favored group's curve.
- When the disadvantaged group is believed to have greater variance in latent promise, the prediction curve can become steeper and may cross the favored group's curve.
- When test scores are believed to be less reliable for a group, the employer places less weight on those scores.
- The paper's key warning is that discrimination can be damaging even when it is statistical rather than taste-based.

11 Conclusion

11.1 Core conclusion

Phelps shows that discrimination can arise from imperfect information and statistical inference. An employer who has no distaste for a group can still make group-contingent hiring predictions if the employer believes that group membership conveys information about unobserved productivity or about the reliability of the observed signal.

11.2 Economic intuition

The core intuition is Bayesian shrinkage. Employers combine individual information with group-level priors. If individual signals are noisy, group priors matter more. If the employer believes that a group has a different mean, variance, or signal reliability, then applicants with the same observed test score can receive different predicted qualifications.

11.3 Incremental contribution revisited

The paper's lasting contribution is to formalize statistical discrimination in a simple model. It opens a path for later research on discrimination without animus, noisy signals, employer learning, and self-confirming stereotypes.

11.4 Policy implication

The model implies that anti-discrimination policy cannot focus only on overt hostility. It must also address information structures, signal quality, employer priors, and the social processes that make group identity appear predictive.

11.5 Limitations

The model is intentionally simple. It does not estimate magnitudes, model worker skill investment, or derive a dynamic equilibrium of belief formation. It also takes employer priors as given. These omissions limit empirical interpretation but keep the core theoretical mechanism transparent.

11.6 Takeaway

The enduring takeaway is that statistical discrimination can be rational from the perspective of an individual employer while still being unfair, inefficient, and harmful from a social perspective. The mechanism matters because it shows why eliminating animus alone may not eliminate discriminatory outcomes.